REV. FR. NDUKA ANTHONY, C.S.Sp

Priest | Data Scientist | Healthcare Technologist | Researcher in AI & Ethics

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🎓 EDUCATION

PGD in Information Technology **July 2022 – April 2024**
National Open University of Nigeria, Abuja **(**NOUN)
Thesis Title: "Development of INEC Registration and Voting System using Distributed Database"
Supervisor: Dr. Ezema Chukwuedozie

M.A in Theology **July 2018 – June 2019**
Spiritan International School of Theology, Enugu (SIST) **CGPA*:*** 3.47/4.0 (89%)
(Affiliate of Duquesne University of the Holy Spirit, USA)

B.A in Religion and Cultural Studies **Sept. 2015 – June 2018**
Spiritan International School of Theology, Enugu (SIST) **CGPA*:*** 4.45/5.0 (89%)
Thesis Title: "Cyber Evangelization: Exploring the Potentials of Social Media in the Proclamation of the Gospel in Nigeria" *(Grade A)*
Supervisor: Fr. Dr. Matthew Mmuo, CMF

B.A in Philosophy **Sept. 2011 – June 2015**
University of Nigeria, Nsukka. (UNN) **CGPA*:*** 4.18/5.0 (82%)
Thesis Title: "Critique of the Aesthetic Judgment in Kant" *(Grade A)*
Supervisor: Prof. Christopher Asogwa

RESEARCH INTEREST

Data Science for Social Impact | Healthcare Analytics | Ethical AI | Human-Computer Interaction | Big Data & Distributed Systems

RESEARCH EXPERIENCE

Principal Investigator (PDG Project), National Open University of Nigeria **July. 2023 – Dec. 2023**

- Conducted a comprehensive research project on developing a secure and efficient voting system using distributed databases, focusing on enhancing electoral processes.
- Utilized qualitative and quantitative research methods to develop a scalable system capable of handling large volumes of data and concurrent user access, optimizing performance during peak voting periods.

Principal Investigator, (B.A Project) Spiritan International Sch of Theology, Enugu **Jan. 2018 – Apr. 2018**

- Conducted surveys and interviews with over 620 religious leaders and social media users to gather qualitative and quantitative data on the effectiveness of online evangelization, using data visualization skills.
- Used social media and networking analysis to identify key opportunities and challenges in utilizing social media for gospel proclamation, offering recommendations for religious organizations aiming to expand their digital presence.

Principal Investigator, (B.A Project) University of Nigeria, Nsukka **Jan. 2014 – May. 2014**

- Critically evaluated Kant's arguments, identifying strengths and limitations, and proposing alternative interpretations or enhancements to his aesthetic framework.
- Demonstrated strong analytical and communication skills through the presentation of research findings at academic seminars and in written reports.

🛠 TECHNICAL PROJECTS

Clean and Analyze Employee Exit | View **April – May 2024**

- Cleaned and prepared exit survey data from the Department of Education, Training and Employment (DETE) and the Technical and Further Education (TAFE) body, including handling missing values and correcting inconsistencies which improve the Department.
- Utilized Python libraries such as Pandas and NumPy for data manipulation, aggregation, and feature extraction from the survey datasets and conducted exploratory data analysis (EDA) to uncover trends and patterns in employee turnover, identifying key factors influencing employee exits.

Analysis of the Prosper Marketplace – a Loan Company | View **May – June 2024**

- Identified critical business needs by analyzing the loan dataset, addressing the necessity to understand factors influencing loan default rates to improve risk management and lending practices. This informed strategic decisions, resulting in an estimated reduction in default rates by 15%.
- Developed data visualizations using Matplotlib and Seaborn to effectively communicate the impact of borrower demographics and loan terms on repayment success, leading to actionable insights for optimizing loan approval criteria.

Predicting Cardiovascular Diseases | View **July – August 2024**

- Developed a predictive model to classify patients as high or low risk for cardiovascular disease using a dataset from the UCI Machine Learning Repository, addressing the critical need for early detection in healthcare, achieving a classification accuracy of 86.96%.
- Performed hyperparameter tuning using GridSearchCV to enhance model performance, improving the prediction of high-risk cardiovascular cases.
- Collaborated with cross-disciplinary teams to ensure the model's relevance and applicability in real-world healthcare scenarios, emphasizing the importance of data-driven decisions in patient care, while paying attention to issues around ethics.

PUBLICATIONS

- Nduka, A. (2024). "*Real-Time Edge Analytics for IoT Networks: Optimizing Data Processing and Decision-making in Smart Cities*". **DOI: https://doi.org/10.51584/IJRIAS.2024.909012**
- Nduka, A. (2024). "*Leveraging Big Data and IoT to Revolutionize Content Creation, Distribution, and Audience Engagement in the Media Industry*". (*Under Review at* **Science and Technology Publishing, ISSN: 2632-1017**
- Nduka, A. (2024). *"Quantum Computing in Healthcare: Optimization of Systems for Drug Discovery, Personalized Medicine, and Diagnostics".* **(***Under Review at* **International Journal of Innovative Science and Research Technology)**

ACADEMIC PRESENTATIONS

- *"Cyber Evangelization in the Digital Age,"* **NACATHS National Conference**, Enugu – 2018
- *"Spirituality and AI: Harmonizing Data with Mission,"* **Spiritan Provincial Symposium,** Onistha – 2024

TEACHING AND MENTORING EXPERIENCE

Graduate Teaching Assistant, Department of Computer Science, (SUN) **Mar 2024 – Present**

- Assisted undergraduates in programming and data analytics courses to a class size of 15 students.
- Graded and recorded semester assignments and papers, providing constructive feedback to students.

Academic Dean's Secretary, Bishop Okoye Spiritan Secondary School, PH (BOSSS) Sept 2014 – June 2015

- Effectively mentored, invigilated and recorded college students' examinations and results.
- Served as the chair of the board of competition committee, where I prepared 10 students for the 2015 Cowbell Mathematics Competition and one of the students made it to the national level.

Teaching Instructor, Sample Child Secondary School, Aba, Abia State **2008 – 2009**

- Taught a class of about 68 senior high school students mathematics and computer science, and they recorded over 88% success rate in their Senior Secondary Certificate Examination.

💼 PROFESSIONAL EXPERIENCES

Spiritan Press Enterprise Onitsha, Nigeria *(Assistant Director)* **Sept. 2019 – Sept. 2023**

- Supervised the operations of different branches of the centre, with a total staff strength of 28.
- Served as Quality Control manager of both print and digital productions, that raised our customer base by 40%.
- Managed the enterprise's data analytics for optimal operation which raised the centre's annual revenue by 25%.

Spiritans, Province of Nigeria, South East, Nigeria *(Asst. Provincial Secretary*) **2019 – Present**

(The largest Province in the Congregation of the Holy Spirit (Spiritans) worldwide)

- Assists the Provincial Secretary in the daily running of our Provincial Secretariat.
- Manage, analyze and protect the Database of the Province, which led to an official recognition by Rome as the best Provincal data manager since in 2022.
- Serves as the author of the Provincial Biannual Newsletter, serving over 2000 members and associates.

Directorate of Social Communication, Spiritans *(Director)* **October 2023 – Present**

- Leads a multidisciplinary team of 20+ professionals in executing comprehensive social communication strategies which has raised Catholic religious awareness online by over 45%.
- Design and execute data collection and analysis protocols to measure the impact of communication initiatives and drive continuous improvement for various organizations, and so far 8 of such organizations have improved.
- Conducts training sessions for various religious groups on best practices in social communication and media relations, and has spoken to over 2500 persons so far.

Spiritan Healthcare Apostolate – HECAP ***(Founder/Director)*** **October 2022 – Present**

- Designed predictive health models to support elderly care across 4 dioceses in Nigeria.
- Digitized medical records and improved care coordination for retired priests in the Spiritan Congregation.
- Develop predictive models to anticipate healthcare needs and interventions for elderly people, enhancing proactive care management which has assisted over 155 elderly in four dioceses.

VOLUNTEERING & COMMUNITY SERVICE

- **Justice, Peace and Integrity of Creation** **2020 – Present**
  Organizer, Christmas Feeding and Medical Outreach – reaches over 900 persons annually
- **Laudato Si Movement – Vatican City, Rome** **2022 – Present**
  Serve as an active campaigner for averting climatic crises and promoting a greener environment.

LEADERSHIP

President, Theology Department, Spiritan International Sch of Theology, Enugu **Oct. 2018 – May 2019**

- Presided over 320 students in the department, coordinating other executive board members.
- Represented the association at national conventions and won two competitions in the national body of Seminaries.
- Organized Theological symposia and seminars for the School.

President, Philosophy Department, SSPI, University of Nigeria, Nsukka **Oct. 2014 – May 2015**

- Presided over 400 students in the department, and facilitated student/staff relationships
- Represented the association at national conventions and became NationalVice President of NAPS (National Association of Philosophy Students)
- Organized Philosophical weekend get-togethers that brought over 12 different Universities together.

🎖 AWARDS AND HONOURS

- **Vocations Movement Scholarship Award for Seminary Training** **2015 - 2019**

- **Best Paper Presenter** at the National Association of Catholic Theology Students (NACATHS), held at Bigard Memorial Seminary, Enugu. **2018**
- **Most Creative Student of the Year** at Spiritan International School of Theology (SIST). **2019**
- **Most Efficient Data Management Secretary**, award by the Spiritan Generalate, Rome **2022**

SKILL SUMMARY

- **Language:** Python, R, SQL
- **Framework:** Pandas, Numpy, Matplotlib, Scikit-Learn
- **Tools:** Excel, PowerPoint, Power BI, Tableau, MySQL, SQLite
- **Platform:** Jupyter Notebook, Visual Studio Code, Pycharm
- **Soft Skills**: Excellent communication, Team building, Professional Networking, People Management

CERTIFICATIONS

- **Data Science Professional Certificate** (IBM) | Certificate **August 2023**
- **Data Science Ethics** – University of Michigan | Certificate **October 2023**
- **Data Analysis using Python** (ACETEL) **March 2023**
- **Introduction to Data Science** (IBM) | Certificate **August 2023**
- **Introduction to Statistics** – Stanford University | Certificate **September 2023**
- **Business Analytics with Excel** – John Hopkins University | Certificate **October 2023**
- **Data Analytics** Bootcamp – (Lighthall) **February 2023**

PROFESSIONAL AFFILIATIONS

- Data Scientists Network (DSN) - Member since 2020 (ID: DSN/AIPlus/2020/35228) **2020 – Present**
- NACATHS member